COHOMOLOGICAL BOUNDS FOR A FUNCTOR OF FINITE COHOMOLOGICAL DIMENSION APPLIED TO A QUASI-COHERENT COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The negative vanishing property of derived functors under the canonical truncation. invoked at proof lines 395-397. Supplies statement (1) directly.
- For a K-injective resolution the functor applied to the complex agrees with the derived limit of its truncations, and the cohomology of a derived limit is computed by the Rlim short exact sequence. invoked at proof lines 400-410. Expresses H^i(RF(E)) via the inverse system of the truncated values.
- The truncation triangles produce maps between the truncated systems whose cohomology maps are isomorphisms past the cohomological dimension bound, giving the ML condition and eventual constancy. used at proof lines 414-437. Shows the R^1lim term vanishes and the limit term stabilizes to the value at a finite truncation.

Source reference. Original lemma 05TC: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 395-397 for statement (1); the remaining statements are proved inside the counted span.

Source reference. Original lemma 08D3: sources/stacks-perfect/pinned-original-source.tex, source lines 353–353. Cited at proof lines 400-402 for E = Rlim E_n; the application to RF and the Rlim computation are inside the counted span.

Source reference. Original lemma 08U1: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 403-405 to get RF(E) = Rlim RF(E_n); the short exact sequence and its analysis are inside the counted span.

Source reference. Original remark 08U5: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 407-410 for the Rlim short exact sequence; the ML argument consuming it is inside the counted span.

Source reference. Original remark 08J5: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 414-419 for the triangle between E_n and E_{n-1}; the isomorphism of the cohomology systems is derived inside the counted span.

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Source reference. Original section 07KV: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 433-436 for the vanishing of R^1lim under the ML condition; the eventual constancy argument is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $F : Mod(\mathcal{O}_X) \to Ab$ be an additive functor and $N \geq 0$ an integer. Assume that*

1. *F commutes with countable direct products,*
2. *$R^pF(\mathcal{F}) = 0$ for all $p \geq N$ and $\mathcal{F}$ quasi-coherent.*

Then for $E \in D_{QCoh}(\mathcal{O}_X)$

1. *$H^i(RF(\tau_{\leq a}E)) \to H^i(RF(E))$ is an isomorphism for $i \leq a$,*
2. *$H^i(RF(E)) \to H^i(RF(\tau_{\geq b-N+1}E))$ is an isomorphism for $i \geq b$,*
3. *if $H^i(E) = 0$ for $i \notin [a, b]$ for some $-\infty \leq a \leq b \leq \infty$, then $H^i(RF(E)) = 0$ for $i \notin [a, b + N - 1]$.*

Proof. Statement (1) is Derived Categories, Lemma 05TC.

Proof of statement (2). Write $E_n = \tau_{\geq -n}E$. We have $E = R\lim E_n$, see Lemma 08D3. Thus $RF(E) = R\lim RF(E_n)$ in $D(Ab)$ by Injectives, Lemma 08U1. Thus for every $i \in \mathbf{Z}$ we have a short exact sequence

$$0 \to R^1 \lim H^{i-1}(RF(E_n)) \to H^i(RF(E)) \to \lim H^i(RF(E_n)) \to 0$$

see More on Algebra, Remark 08U5. To prove (2) we will show that the term on the left is zero and that the term on the right equals $H^i(RF(E_{-b+N-1}))$ for any b with $i \geq b$.

For every n we have a distinguished triangle

$$H^{-n}(E)[n] \to E_n \to E_{n-1} \to H^{-n}(E)[n+1]$$

(Derived Categories, Remark 08J5) in $D(\mathcal{O}_X)$. Since $H^{-n}(E)$ is quasi-coherent we have

$$H^i(RF(H^{-n}(E)[n])) = R^{i+n}F(H^{-n}(E)) = 0$$

for $i + n \geq N$ and

$$H^i(RF(H^{-n}(E)[n+1])) = R^{i+n+1}F(H^{-n}(E)) = 0$$

for $i + n + 1 \geq N$. We conclude that

$$H^i(RF(E_n)) \to H^i(RF(E_{n-1}))$$

is an isomorphism for $n \geq N - i$. Thus the systems $H^i(RF(E_n))$ all satisfy the ML condition and the $R^1 \lim$ term in our short exact sequence is zero (see discussion in More on Algebra, Section 07KV). Moreover, the system $H^i(RF(E_n))$ is constant starting with $n = N - i - 1$ as desired.

Proof of (3). Under the assumption on E we have $\tau_{\leq a-1}E = 0$ and we get the vanishing of $H^i(RF(E))$ for $i \leq a - 1$ from (1). Similarly, we have $\tau_{\geq b+1}E = 0$ and hence we get the vanishing of $H^i(RF(E))$ for $i \geq b + N$ from part (2). □